

# The Provable-Results Portfolio

Eighteen theorems, bounds, and experiments I can carry end-to-end,  
with lift estimates and working drafts  
For *The Harbor, the Person, and the Economy* (Owens, v1.0)

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**Scope and honesty rule.** This document enumerates every result from our exchanges that I judge myself capable of developing *rigorously* — meaning I can state the model, carry the proof or derivation myself, and tell you exactly which assumptions bear the load — as opposed to results I could only sketch or survey. Each entry gives the lift (my estimate of focused working time to a polished, checkable draft), a formalization, the actual proof path with its crux, and the payoff to the volume. A closing section lists what I *decline* to claim, in the volume’s own spirit. Tier A items are days each; Tier B items are roughly a week each; Tier C items are multi-week. Several Tier A items are independent enough to batch.

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## 1 Tier A — days each

### 1.1 A1. The Split-Digest Theorem (comonotonicity characterization)

**Lift:** 1–2 days; the proof is short, the care is in the definitions.

**Formalization.** Define a *read-surface* as a triple  $(X, \Sigma, \pi)$ : substrate items  $X$ , per-item evidence  $\Sigma$ , and a projection  $\pi$  producing a bounded rendering. A *reader*  $r$  is a loss function  $L_r(\text{rendered set}, \text{true states})$ ; a scalar head  $h : X \rightarrow \mathbb{R}$  *serves*  $r$  if rank-ordering by  $h$  realizes  $r$ ’s optimal rendering at every budget. Two concrete readers: the *successor* (loss = task-continuation value forgone; fit-shaped) and the *operator* (loss = regret-if-ignored; risk-shaped). The theorem: a single head serves both readers *iff* their induced preference orders are comonotone on the item domain — i.e., agree on every pair. The negative direction is witnessed by two constructible items: a routine-but-essential working state (high continuation value, negligible regret — the successor must reload it; the operator need never see it) and an anomalously abandoned dead end with irreversible side effects (zero continuation value — the successor should *not* reload it — maximal regret). Non-comonotone; hence no shared head; hence Thm. I.7.1 is a special case of a general principle over every read-surface: digests, checkpoints, resurrection handoffs, briefings.

**Proof path.** The order-reversal argument is your Thm. I.7.1’s, generalized: strict monotone transforms preserve pairwise order, so a shared head forces comonotonicity; conversely, comonotone orders admit a common refinement, which any strictly increasing reparameterization serves — giving the *iff*, which the original theorem lacks. The quantitative addendum is where new content lives: apply Thm. I.6.1 per reader. Each reader has its own critical set  $S_r$ ; a *joint* zero-miss digest must satisfy the floor for  $S_{\text{op}} \cup S_{\text{succ}}$ , and when the sets are nearly disjoint the joint floor approaches the *sum* of the individual floors — a single stored compaction pays both bills at once or silently defaults on one. Corollary: compaction must be computed per reader *from durable artifacts*; storing only one compaction is irreversible information loss for the other reader, which is exactly the “never compact from the previous summary” rule, now with a reason.

**What it gives.** A one-page theorem that converts an architectural opinion (the SLM sidecar; two digest pipelines) into a mathematical necessity, upgrades Thm. I.7.1 from an example to a characterization, and quantifies the cost of ignoring it via additive bit floors. Lands in Chapter I as §I.8.2’s missing second half.

### 1.2 A2. Deriving the regret head (and the surfacing threshold) from decision theory

**Lift:** 1–2 days.

**Formalization.** The volume asserts  $\text{regret}(x) = \text{stakes}(x) \times \text{irreversibility}(x) \times \text{anomaly}(x)$  as a design choice. It is derivable. Let  $C_{\text{miss}}(x) = \text{stakes}(x) \cdot \text{irreversibility}(x)$  be the realized loss if  $x$  is bad and unattended, and let the anomaly score be a *calibrated posterior*  $\Pr(\text{bad} \mid \text{evidence about } x)$ . Then expected loss of not surfacing is exactly  $C_{\text{miss}}(x) \cdot \Pr(\text{bad} \mid x)$  — the multiplicative form, no longer asserted but exact, *iff* anomaly is calibrated probability rather than an arbitrary score.

**Proof path.** Direct expected-cost minimization: surface  $x$  iff  $C_{\text{miss}}(x) \Pr(\text{bad} \mid x) \geq c_{\text{att}} + C_{\text{fa}} \Pr(\text{good} \mid x)$ , where  $c_{\text{att}}$  is the operator’s marginal attention cost. Rearranged, this is the classical SDT likelihood-ratio criterion  $\beta = \frac{1-\pi}{\pi} \cdot \frac{C_{\text{fa}}}{C_{\text{miss}}}$  that Def. I.5.1 gestures at — now derived rather than cited, with the queue’s forced-zoom rule falling out as the optimal policy. Two corollaries do real work: (i) *miscalibration correction* — if the raw anomaly score  $s$  is not calibrated, the correct head is  $C_{\text{miss}} \cdot g(s)$  with  $g$  the empirical calibration map, fittable from the audit log (a measurable object, per Appendix B.3.6); (ii) the Goodhart surface sharpens — agents attack  $g$ , not the formula, so the defense budget belongs on calibration integrity, unifying with A6/B2’s inspection accounting.

**What it gives.** Equation I.3 stops being a heuristic. The attention queue becomes an optimal alerting policy with a named optimality criterion, a calibration obligation, and a place where the empirical program (instrument false alarms and misses first) plugs directly into the math.

### 1.3 A3. $\varepsilon$ -conservation for the clean-room release ledger

**Lift:** 1 day, plus imports.

**Formalization.** Model the release ledger as a state machine: state = (cumulative spend  $\sigma$ , append-only log  $\Lambda$ ); the only spending transition is  $\text{release}(\varepsilon_i)$ , which atomically appends  $(\varepsilon_i, \text{artifact hash}, \text{policy hash})$  and sets  $\sigma += \varepsilon_i$ ; the gate refuses when  $\sigma + \varepsilon_i > \varepsilon_{\text{max}}$ .

**Proof path.** Induction on the transition sequence gives  $\sigma = \sum_{\Lambda} \varepsilon_i$  invariantly — the identical proof shape as the bond-conservation theorem (VI.10.1), including the concurrent-invocation lemma (no over-spend under two racing releases, by the single-writer serialization the kernel already provides). Then import composition: sequential composition (budgets add) for the baseline, advanced composition ( $\varepsilon' = \sqrt{2k \ln(1/\delta')} \varepsilon + k\varepsilon(e^\varepsilon - 1)$ ) when the engagement is long and the tighter accounting is worth its bookkeeping. The one honest caveat to state in-line: conservation governs *recorded* spend; the claim that all spend is recorded is exactly the complete-mediation assumption (B3), which is why the hypervisor is a prerequisite, not a nicety.

**What it gives.** The clean-room’s headline invariant, proved with machinery the volume already owns — a structural rhyme (the harbor conserves one more scarce quantity through buckets) that makes the Derek/Erin product auditable rather than promissory.

### 1.4 A4. Canary detection power and sequential leak testing

**Lift:** 1 day.

**Formalization.** Plant  $c$  canaries (unique sentinel spans) uniformly in Derek’s corpus. The gate’s verbatim/embedding suppressor has per-canary false-negative rate  $\beta$ . A leak carrying  $k$  canary-bearing spans is detected with probability  $\geq 1 - \beta^k$ ; solve for the canary density needed so that a leak of economic size  $V$  carries  $k(V)$  canaries in expectation, giving a published operating curve  $\Pr(\text{detect}) = f(V)$ .

**Proof path.** Independence across canaries under uniform planting gives the  $1 - \beta^k$  bound directly; the refinement worth writing is the *sequential* version — Wald’s SPRT on the stream of gate outputs against the permutation null (no-leak), yielding expected time-to-detection as a function of leak rate, with the standard SPRT optimality property. This turns “we have canaries” into “here is the detection-power curve and the expected latency at each leak intensity,” the same move Chapter VI makes for bonds.

**What it gives.** Assurance as a purchasable quantity on the leakage side: the clean-room contract can *quote* detection probability, and the exfiltration bond (which cannot cover unbounded breach damage) is re-scoped to the honest job of funding detection and response — the layer ordering the volume already believes, now with numbers.

## 1.5 A5. The directory-routed message load bound

**Lift:** 2 days.

**Formalization.** Agents generate messages at rate  $\lambda$ ; relevance of a message to agent  $i$  follows a concentration law (Zipf with exponent  $\alpha > 1$  over ranked recipients is the natural model, since most coordination messages concern few parties). Broadcast delivers  $\Theta(n\bar{s})$  tokens per agent per epoch. Directory-routed delivery with fan-out cap  $f$  delivers  $O(f\bar{s})$ ; the theorem quantifies the recall price: to guarantee expected relevant-message recall  $\geq 1 - \epsilon$ , the required fan-out is  $f^* = \Theta(\epsilon^{-1/(\alpha-1)})$  — a constant in  $n$ .

**Proof path.** Tail-mass computation on the Zipf law (the mass beyond rank  $f$  is  $\Theta(f^{1-\alpha})$ ; invert). The honest bookkeeping: the  $\log n$  in my earlier note belongs to directory *lookup* cost, not delivered load; separate the two lines. Add the adversarial rider from the review: since inbound messages are attacker-influenced context, the bound should be stated over *provenance-labeled* tokens, and the injection defense (quoted-channel rendering) costs a constant factor, not asymptotics.

**What it gives.** Chapter II’s bus gets its missing scaling claim: messaging that stays  $O(1)$  in swarm size at fixed recall, with the recall/fan-out dial exposed — the read-poverty thesis extended from discovery to communication, in one page.

## 1.6 A6. Reputation-inheritance conservation for distilled skills (the no-mint theorem)

**Lift:** 2–3 days; the design choice is the content.

**Formalization.** Let witnessed outcomes form the leaves of an evidence DAG whose internal nodes are derived artifacts (CDM-distilled skills, revisions, merges). Assign each outcome  $e$  a graded value  $q(e)$  and a *derivation budget*: weights  $w(e \rightarrow a) \geq 0$  over all artifacts  $a$  derived from  $e$ , constrained by  $\sum_a w(e \rightarrow a) \leq 1$ . An artifact’s inherited prior is  $\iota(a) = \sum_e w(e \rightarrow a) q(e)$ , recursively through the DAG with budgets consumed at each hop.

**Proof path.** The theorem: total inherited prior over the DAG’s closure never exceeds total witnessed value,  $\sum_a \iota(a) \leq \sum_e q(e)$ , and no distillation cycle mints value. Proof by a potential function — each outcome carries at most unit budget, each derivation strictly consumes it, so the sum is a supermartingale under any sequence of distillations; cycles are safe because re-derivation from an artifact draws on the artifact’s (already-discounted) budget, not the leaf’s. The crux is the design decision the proof forces into the open: budgets are *per outcome, split across all derivations*, not per derivation — otherwise the distill-loop mints arbitrarily (derive  $k$  skills from one episode, each inheriting full value).

**What it gives.** Closes Gap #4 (skill versioning cold-start) safely: v2 warm-starts from provenance, fraud cannot amplify through distillation chains, and the CDM pipeline (Mechanism 6) gets its economic safety proof. Also the estate’s export/import ceremony inherits the same conservation for free.

## 1.7 A7. The Theorem I.6.1 falsification experiment (executable)

**Lift:** hours to a first figure; 1–2 days to a clean writeup — I can run this in the container on request.

**Formalization.** Simulate  $N$  artifacts with a hidden critical set of size  $k$ ; construct digests at token budgets  $B$  sweeping through the information floor  $\log_2 \binom{N}{k} - \log_2 \binom{m}{k}$ : (i) an optimal covering-design digest at small  $N$ , (ii) greedy set-cover digests, (iii) a learned scorer over noisy features. Measure empirical miss rate versus  $B$  and overlay the floor.

**Proof path.** Not a proof — a falsification apparatus. The prediction is sharp: zero-miss is unattainable below the floor for *any* digest scheme, and the empirical curves should hug it from above, with the gap between greedy/learned and the floor measuring the price of realistic (non-oracle) digest construction. The second panel is the adversarial variant from Solution 6.5\*: corrupt a fraction  $\phi$  of features and watch the effective floor inflate — the first empirical trace of the mimicry/inflation attack. The third panel closes the loop with A1: measure the joint-digest penalty when two readers’ critical sets are disjoint versus overlapping.

**What it gives.** The volume’s first datum, its best figure, and a public artifact (the beginning of the coordination benchmark the review argued you should own). It also stress-tests the theorem’s modeling: if real digests systematically beat the floor, the critical formalization is wrong, and finding that out cheaply is the point of having stated a falsifiable bound.

## 2 Tier B — roughly a week each

### 2.1 B1. The digest–zoom Pareto frontier: two-stage group testing and one-sided rate–distortion

**Lift:** 1 week; the derivations are standard in technique, the assembly is new.

**Formalization.** The digest writer knows the state  $X^N$  ( $X_i \in \{\text{critical}, \text{inert}\}$ , i.i.d. Bernoulli( $p$ ) in the average-case model); the operator does not. A digest is  $B$  bits about  $X^N$ ; the operator opens the flagged set and may then *zoom* adaptively. Total operator cost:  $c_{\text{read}}B + c_{\text{open}}\mathbb{E}|\hat{S}| + c_{\text{miss}}\mathbb{E}|S \setminus \hat{S}|$ . The right information-theoretic object is a *two-constraint* rate–distortion program:

$$R(\delta, f) = \min_{P_{\hat{X}|X}} I(X; \hat{X}) \quad \text{s.t.} \quad \Pr(X=1, \hat{X}=0) \leq \delta, \quad \Pr(\hat{X}=1) \leq f,$$

false negatives priced by  $\delta$  (misses), false positives priced only through the open budget  $f$ . Your Thm. I.6.1 is the ( $\delta=0, f=m/N$ ) worst-case corner of this program.

**Proof path.** Three deliverables. (i) Solve  $R(\delta, f)$  in closed form for the binary source via the Lagrangian/test-channel method — a finite convex program whose KKT conditions I can carry by hand; verify the corners ( $f=1 \Rightarrow R=0$ : flag everything;  $\delta=0, f=p \Rightarrow R \rightarrow h(p)$ -scale: the digest must localize). (ii) The *zoom advantage theorem*: adaptive second-stage inspection with group-level summaries (zooming a directory before its files) is Hwang-style generalized binary splitting inside the flagged region, reducing expected opens from  $fN$  to  $k \log_2(fN/k) + O(k)$  whenever group verification is available — quantifying, for the first time, what the “zoom” in digest-with-zoom is mathematically worth over a flat digest. (iii) Assemble (i)+(ii) into the Pareto frontier (digest bits)  $\times$  (expected opens) at fixed miss tolerance, with the operator’s optimum at the price-ratio tangent point  $c_{\text{read}}/c_{\text{open}}$ .

**What it gives.** Chapter I’s §I.6 grows from one worst-case bound into a small theory: average-case floors, the formal value of adaptivity, and a knob (the tangent point) that turns operator time-accounting into digest sizing. It also gives A7’s experiment its second axis. Of everything here, it is the strongest single candidate for a standalone paper — the mathematical identity of “summarize the state, link to the work.”

## 2.2 B2. The inspection tower: critical audit rates, contraction, and reputation as amortized verification

**Lift:** 1–1.5 weeks.

**Formalization.** Level  $k$  is an inspection game: the judge cheats with probability  $q$  for one-shot gain  $G_k$ ; the inspector audits with probability  $\rho_k$  at cost  $a$  per audit, detecting with probability  $d_k$  and slashing bond  $B_k$ . Levels stack: level  $k+1$  audits level- $k$  auditors, sampled sealed from a pool spanning  $\geq q_{\text{cl}}$  disjoint principal-cliques. Above the tower sits a reputation model in the Kreps–Wilson/Fudenberg–Levine mold: a fraction  $\mu_0$  of judges are commitment (honest) types; strategic types decide whether to mimic.

**Proof path.** Four steps, each standard in technique. (i) The stage game’s committed-inspection solution: deterrence at  $\rho_k^* = G_k/(d_k B_k)$ , minimal audit spend  $a G_k/(d_k B_k)$  per judged outcome — Becker’s condition, now with the audit *cost* accounted, which the conjecture omits. (ii) Contraction: under sealed sampling from  $q_{\text{cl}}$  cliques, a briber corrupting level- $k+1$  must in expectation pay off order  $q_{\text{cl}}$  independent auditors’ expected forfeitures before capturing the audit, so  $G_{k+1} \leq G_k \lambda$  with  $\lambda = O(1/q_{\text{cl}})$ ; then required bonds  $B_k \gtrsim G_0 \lambda^k / (\rho d)$  sum geometrically — *finite total bond capitalization certifies an infinite tower*, which is the contraction Conj. III.11.1 asks for, with its two assumptions (non-shrinking stakes, sealed multi-clique sampling) named as the load-bearers. (iii) Termination alternative: a finite tower rooted at an exogenously honest oracle (the operator at  $n=1$ ) with the same inequality at each level. (iv) The reputation layer: derive the declining audit schedule  $\rho_t$  that keeps the strategic type’s IC binding as the posterior  $\mu_t \rightarrow 1$  — audits fall as continuation value rises — and prove lifetime audit spend per honest judge is  $O(1)$  versus  $\Theta(T)$  under flat auditing. That last result *is* “reputation is amortized verification,” stated as a theorem.

**What it gives.** The volume’s central open conjecture becomes a parameterized theorem plus a measurement obligation (does the deployed pool satisfy sealed multi-clique sampling?),

the audit budget becomes a designed quantity rather than a hope, and Chapter III gains its deepest one-liner with a proof behind it.

### 2.3 B3. Hypervisor enforceability is Ramadge–Wonham controllability

**Lift:** 1 week.

**Formalization.** Model agent behavior as a prefix-closed language  $L$  over an alphabet  $\Sigma = \Sigma_c \cup \Sigma_u$ : *controllable* events  $\Sigma_c$  are those the hypervisor mediates pre-commit (fs writes, network egress, exec), *uncontrollable* events  $\Sigma_u$  are everything it cannot intercept (model-internal steps, in-context content). A policy is a safety specification  $K \subseteq L$ .

**Proof path.** Direct adaptation of supervisory control theory: a supervisor achieving exactly  $K$  exists iff  $K$  is controllable —  $\overline{K}\Sigma_u \cap \overline{L} \subseteq \overline{K}$ , i.e., no uncontrollable event escapes the specification’s closure. The theorem for the volume: **Port Daddy’s hypervisor can regiment policy  $P$  iff  $P$  is a prefix-closed safety language controllable with respect to  $\Sigma_u$** ; everything else is detect-and-compensate, by your own Thm. II.8.1. The proof is the Ramadge–Wonham construction specialized to the daemon’s mediation map, plus one wrinkle I will treat honestly: policies referencing unmediated *state* (not just events) need the partial-observation extension (observability/normality conditions), which strictly shrinks the regimentable set and should be stated as such. The corollary is a table the ledger has been missing: force-push (network event) — regimentable; out-of-scope write, deleting a failing test (fs events over policy-expressible paths) — regimentable; “confident lie in a report” (semantic content, uncontrollable) — forever detect-and-compensate. Re-grade every affected ledger row against the alphabet.

**What it gives.** The hypervisor stops being an engineering preference and becomes a theorem about which promises the product may make: an exact, citable boundary between *regimented* and *enforced*, resolving §I.4.4’s open problem (footgun guards as structural authority) for every policy inside the controllable set — and honestly fencing what stays outside it.

### 2.4 B4. A tractable deontic-conflict fragment for the condominium harbor

**Lift:** 1 week.

**Formalization.** Commitment language  $\mathcal{L}_c$ : (a) ground Horn clauses over a finite atom vocabulary (invariants, accepted obligations); (b) interval resource claims (res,  $[t_1, t_2]$ , owner, excl); (c) difference constraints linking commitment deadlines. A *conflict* is a deontic inconsistency: derivable  $\perp$ , overlapping exclusive intervals, or an infeasible constraint cycle.

**Proof path.** Three results. (i) Membership: consistency in  $\mathcal{L}_c$  is decidable in polynomial time and *incrementally* — Horn by watched-literal unit propagation (linear in occurrence size per update), intervals by interval trees ( $O(\log n)$  per insert), difference constraints by negative-cycle detection on the constraint graph; I will write the combined incremental algorithm and its amortized bound. (ii) The frontier: one step outside — arbitrary clauses, or disjunctive obligations  $O(a \vee b)$  — makes consistency NP-complete (reduction from SAT, immediate), so the language design is not taste but the exact price of staying detectable-at-proposal-time. (iii) Completeness-with-regimentation: within  $\mathcal{L}_c$ , every conflict is detected

before acceptance, and since acceptance is a mediated write, B3 makes conflicting acceptances *regimentable* — rejected pre-commit, extending Thm. VI.5.1’s advisory completeness to structural prevention in the shared-harbor setting.

**What it gives.** The condominium chapter’s algorithmic core: the lookout’s contradiction-scan becomes a polynomial incremental checker with a proved completeness property, “who decides what conflicts are” splits cleanly into mechanical detection and chartered resolution, and the schema designer gets a theorem telling them exactly which expressive conveniences would forfeit tractability.

## 2.5 B5. Engine substitution: Akerlof unraveling inside one identity, and resurrection soundness

**Lift:** 4–6 days.

**Formalization.** Two engine qualities  $\theta_H > \theta_L$  with costs  $c_H > c_L$ ; an identity carries reputation  $r$  earned under some engine history; buyers observe  $r$  but, absent attestation, not the current engine. A strategic principal may substitute  $\theta_L$  after building  $r$  on  $\theta_H$ .

**Proof path.** Two-period model, standard lemons technique. Without attestation, buyer beliefs price the identity at  $\mathbb{E}[\theta \mid r, \text{equilibrium strategy}]$ ; substitution is profitable at any pooled price above  $\theta_L$ ’s cost, so pooling on  $\theta_H$  cannot survive: either prices discount to the mixture (taxing honest  $\theta_H$  sellers, who exit or substitute — the ratchet to  $\theta_L$ ) or the market unravels — Akerlof’s argument run *within* a single identity, which is the novel framing. With daemon-attested engine identifiers on every witnessed outcome, reputation conditions on (person, engine class), the information asymmetry closes, and I then prove *resurrection soundness*: under (i) credential and continuity-witness verification, (ii) engine attestation recorded in successor outcomes, and (iii) open commitments closed or renegotiated with escrow delta, the sanction-respecting property (Def. III.6.1) is preserved across provider migration — the migration cannot be used to shed either sanctions or engine history. The same theorem, read sideways, closes skill-versioning Gap #4: reputation attaches to (identity, substrate) pairs with disclosed substrate, one rule for both holes.

**What it gives.** The resume-button feature ships with its own security theorem and its own named attack; Chapter III gains the market-microstructure result it currently lacks; and the buyer-facing story becomes crisp: reputation you can trust *because* the engine behind it is attested.

## 2.6 B6. The newcomer ramp as screening: the front-loaded probation theorem

**Lift:** 3–5 days; the result is a short exchange argument with a surprising sign.

**Formalization.** Two types enter as newcomers: honest (discount  $\delta_h$ ) and fraud-planning (effective discount  $\delta_f < \delta_h$ , since their horizon truncates at the planned reset). The designer chooses the ceiling schedule  $\{\kappa_t\}$ , equivalently the per-period surplus gaps  $g_t = \pi(\kappa^*) - \pi(\kappa_t) \geq 0$ . Deterrence requires the *fraudster’s* discounted ramp cost to exceed the maximal one-shot gain:  $\sum_t \delta_f^t g_t \geq G_{\max}$ . The design objective: minimize the *honest* newcomer’s discounted burden  $\sum_t \delta_h^t g_t$  subject to that constraint.

**Proof path.** The ratio of deterrence-value to honest-cost of a unit gap at time  $t$  is  $(\delta_f/\delta_h)^t$ , strictly decreasing in  $t$ . An exchange argument — move gap mass from any later period to any earlier one, holding the deterrence constraint tight — strictly lowers honest cost; iterate to the corner. Conclusion: the optimal schedule is *maximally front-loaded*: the deepest ceiling restriction immediately, released as fast as the deterrence constraint permits — a probation cliff, not a long slow ramp. The result is mildly counter-intuitive (deferred-compensation instincts, à la Lazear, suggest back-loading) and the resolution is instructive to write out: here the punishment *is* the ramp re-paid after reset, so it must land where the short-horizon type still feels it — early. Extensions I can carry in the same draft: continuum of types (the cliff persists), and the comparative static that as fraud gains  $G_{\max}$  grow, the cliff deepens before it lengthens.

**What it gives.** Theorem III.6.2 currently says “choose the schedule so  $W$  exceeds the fraud gain”; this pins down *which* schedule, with a clean behavioral prediction (probation cliffs dominate gradual ramps) that a deployment can implement tomorrow and A/B against churn.

## 2.7 B7. The costly-escalation signaling equilibrium and the tuning band for the debit

**Lift:** 4–6 days.

**Formalization.** Agents privately observe urgency  $u \sim F$ ; escalating surfaces the item; the operator validates iff genuinely urgent (revealed on attention). A dismissed escalation debits standing by  $\delta$ , worth  $w$  per unit in future surfacing weight; escalating when urgent yields benefit  $b(u)$ , increasing.

**Proof path.** Single-crossing: expected payoff of escalating,  $b(u) - \delta w(1 - V(u))$  with validation probability  $V(u)$  increasing, crosses zero once, so the equilibrium is a threshold  $u^*(\delta)$  solving  $b(u^*) = \delta w(1 - V(u^*))$  — a separating equilibrium in the Spence mold, with the queue’s “crying wolf is costly” realized as money-burning that only high types rationally risk. Comparative statics give the tuning band:  $u^*$  increases in  $\delta$ ; alarm load is  $1 - F(u^*)$ ; false silence (genuine distress suppressed) is  $F(u_{\text{crit}}) - F(u^*)$  truncated at zero. The deliverable is the two-sided constraint  $\delta \in [\delta_{\min}, \delta_{\max}]$ : below  $\delta_{\min}$  the operator’s alarm-fatigue budget (from A2’s attention accounting) is violated; above  $\delta_{\max}$  the miss-risk on the distress lane exceeds the SDT criterion. Both endpoints are computable from instrumented quantities the roadmap already plans to collect.

**What it gives.** Section I.7.3’s mechanism gets an equilibrium existence proof, an explicit operating band for its one free parameter, and a bridge to the empirical program:  $\delta$  is fit from measured validation rates, not chosen by vibes.

## 2.8 B8. The specialization boundary for sole-responsibility roles

**Lift:** 3–4 days.

**Formalization.** Function- $F$  requests arrive  $\text{Poisson}(\lambda_F)$ . Option S: a sole specialist, M/M/1 with service rate  $\mu_s$ . Option P: fold the traffic into a pool of  $c$  generalists, M/M/ $c$  at rate  $\mu_g \leq \mu_s$  each (the specialist’s rate advantage is the skill premium). Sole ownership additionally yields accountability value  $A$  per period (clean attribution, per-role ledger, succession clarity).

**Proof path.** Exact comparison via Erlang-C: closed form for  $c = 2$ , the general condition as an inequality in the Erlang-C delay probability. The theorem is the boundary: sole ownership dominates iff  $\mu_s/\mu_g \geq g(\rho, c)$  or the latency deficit is smaller than  $A$  priced at the delay cost — with  $g$  explicit, and the known queueing folklore (pooling wins on variance) now carrying its exceptions in the open: high skill premia, low utilization, or high attribution value. A worked corner I will include because it matters operationally: the specialist as single point of failure — M/M/1 with breakdowns (the death/heartbeat model the volume already has) — which quantifies the succession-rule requirement rather than leaving it as hygiene.

**What it gives.** Mechanism 2 (calendar avatar, test writer, roadmap owner) gets an honest boundary instead of an aesthetic: when the pattern pays, when it silently costs latency, and what the succession rule is worth — all in quantities the daemon already meters.

## 2.9 B9. Context paging with an untrusted pin oracle

**Lift:** 4–5 days; mostly a disciplined import with one new term.

**Formalization.** Resident context is a cache of size  $k$  tokens over spans in the durable buffer; span requests follow the agent’s action sequence; eviction is online. Critical spans are *pinned* by a feature oracle that an adversary corrupts at rate  $\phi$  (mis-pin or missed pin).

**Proof path.** Import the weighted-caching result (Landlord):  $k/(k - h + 1)$ -competitive against offline with cache  $h$ , applied to the unpinned region — variable span sizes are exactly why the weighted/file-caching version is needed rather than vanilla LRU’s Sleator–Tarjan bound, and I will keep that distinction honest. The new content is the degradation term: a charging argument that mis-pins cost at most their carried size (wasted residency) and missed pins cost at most one re-fetch each, giving total cost  $\leq$  Landlord bound on the honest region  $+ \phi N \cdot c_{\text{refetch}}$  — linear, graceful, and tied by the same  $\phi$  to Solution 6.5\*’s forgeable-feature budget, so the paging layer and the digest layer share one adversarial parameter.

**What it gives.** Mechanism 3’s buffer stops being plumbing: a competitive guarantee for “what stays in context,” a single named adversarial knob shared with the legibility theory, and the formal restatement of the effective-context trap — paying for tokens that competitive analysis says should have been evicted.

## 3 Tier C — multi-week

### 3.1 C1. Conditional noninterference for the clean room via unwinding

**Lift:** 2–3 weeks; standard technique, but the system model must be built with care.

**Formalization.** Domains  $\{D, E, G_{\rightarrow E}, G_{\rightarrow D}, \text{pub}\}$  with an *intransitive* flow policy:  $D \rightsquigarrow G_{\rightarrow E} \rightsquigarrow E$  and  $E \rightsquigarrow G_{\rightarrow D} \rightsquigarrow D$ , with the direct edges  $D \rightsquigarrow E$  and  $E \rightsquigarrow D$  absent. The system is the hypervisor-mediated transition machine: agent steps, gate steps, ledger steps.

**Proof path.** Rushby’s intransitive-noninterference framework: define the purge function for the policy, state the security condition (observations of  $E$  depend only on sources permitted via the gate chain), and discharge it by the three unwinding conditions — output consistency, weak step consistency, and local respect — proved per transition class. Complete mediation (B3) discharges the “no other edges” premise; gate correctness is assumed,

and I will state it as the residual oracle it is (the clean room *relocates* trust into two auditable declassifiers; it does not eliminate it). The honest scope paragraph writes itself and must be written: the guarantee is possibilistic — timing, token-count, and termination channels are outside the model, as is a human reading gate output and remembering it; the theorem bounds *channels*, not minds.

**What it gives.** The Derek/Erin design’s central promise as a theorem with named unwinding obligations — each one a checkable property of a concrete daemon transition — which is precisely the form in which a security claim can later be mechanized (the same ProVerif-adjacent discipline Chapter V already practices, extended from secrecy of keys to flow of labeled data).

### 3.2 C2. Buying your way out of the Myerson–Satterthwaite corner: the minimum viable take-rate

**Lift:** 2 weeks in a stylized type model.

**Formalization.** The harbor is not a bilateral mechanism: it has an insurer/platform side with fee income  $\Phi$  (bond carry, listing, settlement fees). With an outside budget, ex-post efficient trade is implementable (subsidize the VCG deficit); the question MS actually leaves open here is *how much* budget.

**Proof path.** In the canonical uniform-types model, compute the expected VCG deficit of efficient bilateral trade in closed form (a standard exercise I can carry, and one worth re-deriving cleanly rather than citing), then state the theorem: the platform escapes the MS corner iff expected fee income per trade weakly exceeds expected deficit, giving a *minimum viable take-rate*  $\tau^*$  as an explicit function of the value distributions. Corollaries: (i) the take-rate is the price of ex-post efficiency, a sentence Chapter IV can put on a slide; (ii) with the common-value contamination the review flagged (shared grading oracle correlates values), the relevant distortion shifts from MS to lemons, and the same fee budget is better spent on information design — disclosure of reputation vectors — than on subsidies, which I will state as a scoped comparison rather than a grand unification.

**What it gives.** Chapter IV’s most cited impossibility stops being a corner the design nervously respects and becomes a budget line the platform prices — and the analysis says when subsidy is the right instrument and when disclosure is.

## 4 What I decline to claim

In the ledger’s spirit: **gossip convergence under real partitions** is an empirical distribution, not a theorem worth having beyond the stated model (the volume already grades this correctly). **Cross-operator attestation** is a deployment-and-hardware problem; I can specify the TEE/key-transparency evaluation, not prove the gap closed. **Folk-theorem cartel bounds at the federation layer** are provable only in monitoring models stylized past usefulness; I would write the model-selection memo, not claim the bound. **The categorical/sheaf program** beyond the toy gluing model (already solved in the prior document) adds vocabulary, not power, at current scale — I stand by commit-or-cut. And every result above that touches deployment parameters ( $d$ ,  $\beta$ ,  $\phi$ , calibration maps, validation rates) is a theorem *conditional on measurement*; the instrumentation plan of Appendix B.3.6 is the co-requisite, not an afterthought.

## 5 Suggested batching

Week 1: A1 + A2 + A7 (the theorem, its decision-theoretic twin, and the experiment — one coherent legibility package). Week 2: B3 (hypervisor characterization; unblocks the most ledger rows and C1). Week 3: B2 (inspection tower). Week 4: B1 (the frontier paper). Then B5+B6 as the Chapter III pair, B4 for the condominium, and C1/C2 as the long arcs. A1+A2+A7+B1 together are a self-contained paper: *“The Price of a Summary: Information Floors, Adaptive Zoom, and Split Heads for Operator Legibility.”*